

Q1.

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer:

```
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);
    int i = 1;
    int s = 1;
    int count = 2;
    while (1) {
        count++;
        if (!(s <= n))
            break;
        i++;
    }
```

```
    count++;  
    s += i;  
    count++;  
}  
printf("%d", count);  
return 0;  
}
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Q2.

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```

int n;
scanf("%d", &n);
int count = 0;
int i, j;
count++;
if(n!=1)
{
    count++;
    i = 1;
    while (1)
    {
        count++;
        if (i > n)
            break;
        count++;
        j = 1;
        while (1)
        {
            count++;
            if (j > n)
                break;
            count++;
            break;
        }
        count++;
        i++;
    }
}
printf("%d", count-1);
return 0;
}

```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Q3.

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>  
  
int main()  
{  
    int num, i, counter = 1;  
  
    scanf("%d", &num);  
  
    for (i = 1; i <= num; ++i)  
    {  
        counter += 2;  
  
        if (num % i == 0)  
        {  
            counter++;  
        }  
    }  
}
```

```
printf("%d", counter);
```

```
return 0;
```

```
}
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Q4.

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>

void function(int n) {
    int c = 0;

    c++;

    for (int i = n / 2; i < n; i++) {
        c++;

        for (int j = 1; j < n; j = 2 * j) {
            c++;

            for (int k = 1; k < n; k = k * 2) {
                c++;

                c++;
            }

            c++;
        }

        c++;
    }

    c++;
}
```

```

    c++;
    printf("%d\n", c);
}
int main() {
    int n;
    if (scanf("%d", &n) == 1 && n > 0) {
        function(n);
    }
    return 0;
}

```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Q5.

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>

void reverse(int n) {
    int count = 0;
    int rev = 0, remainder;
    count++;
    while (n != 0) {
        count++;
        remainder = n % 10;
        count++;
        rev = rev * 10 + remainder;
        count++;
        n /= 10;
        count++;
    }
    count++;
}
```

```
count++;  
printf("%d\n", count);  
}  
int main() {  
    int n;  
    if (scanf("%d", &n) == 1 && n > 0) {  
        reverse(n);  
    }  
    return 0;  
}
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓